

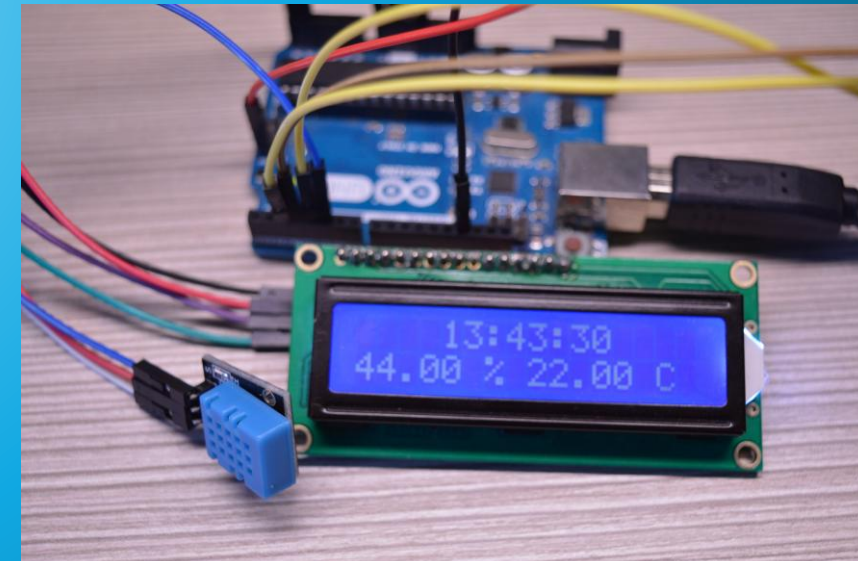
Arduino Weather Station

ECS3001

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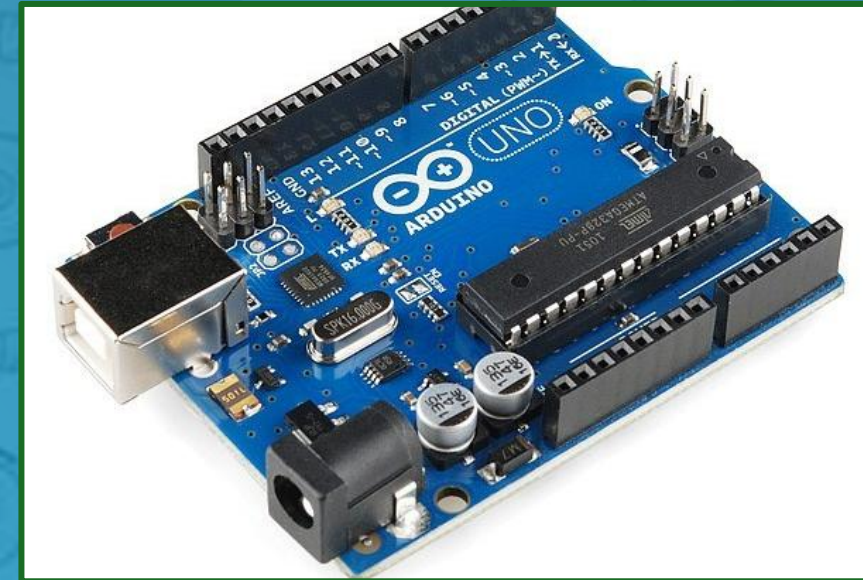


Components

- Arduino Uno
- DHT11 Temperature & Humidity Sensor
- 16×2 LCD Display
- I2C LCD Module
- Jumper Wires
- Breadboard
- Power Supply / Battery

Arduino Uno

The Arduino Uno is the main microcontroller board used to control and process data in the weather station. It receives temperature and humidity readings from the sensor, processes the information using the uploaded program, and sends the results to the display.



Key Features:

- Based on ATmega328P microcontroller
- 14 Digital I/O pins and 6 Analog input pins
- Easy programming using Arduino IDE
- Reliable and widely used for embedded system projects

Application:

- Used as the central controller to read sensor data and control the display of weather information.

DHT11 Temperature & Humidity Sensor

The DHT11 sensor is used to measure environmental temperature and humidity. It provides digital output, making it easy to interface with the Arduino for real-time weather monitoring.



Key Features:

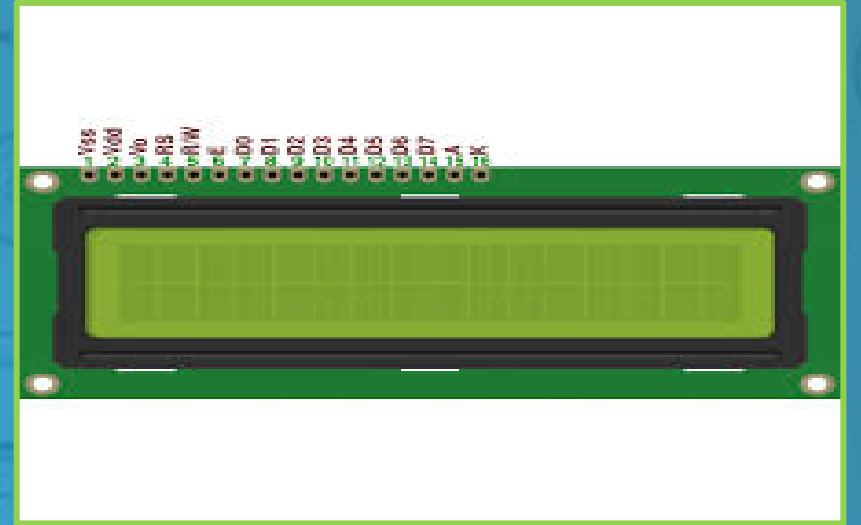
- Measures temperature and relative humidity.
- Digital signal output.
- Low cost and easy to use.
- Suitable for basic environmental monitoring.

Application:

- Used to sense temperature and humidity from the surrounding environment.

16×2 LCD Display

The 16×2 LCD display is used to show the weather data collected by the system. It can display two lines of text, each containing up to 16 characters.



Key Features:

- Displays 2 lines with 16 characters each.
- Low power consumption.
- Clear real-time display of data.
- Compatible with Arduino using I2C or parallel interface.

Application:

- Used to display temperature and humidity readings to the user.

I2C Module

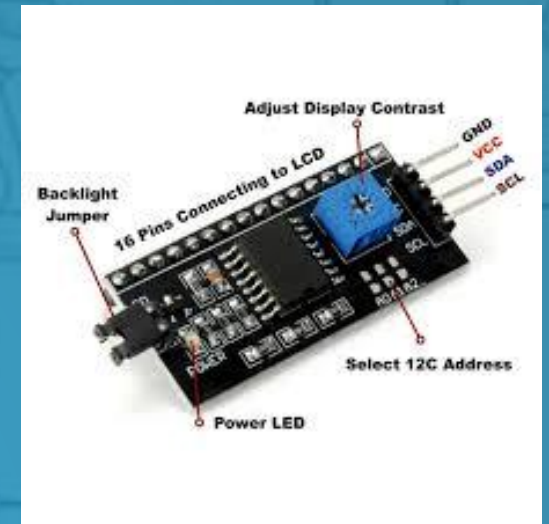
The I2C module is attached to the LCD display to reduce the number of wires required to connect it to the Arduino.

Key Features:

- Uses only two communication lines (SDA and SCL).
- Simplifies wiring connections.
- Supports efficient serial communication.

Application:

- Used to connect the LCD display to Arduino with fewer wires.



Procedure:

Design & Planning

Finalize the project objective and prepare the block diagram. Select the main components such as Arduino Uno, DHT11 sensor, LCD display, and I2C module.



Circuit Assembly

Connect the DHT11 sensor to the Arduino Uno and interface the 16×2 LCD display using the I2C module. Ensure all wiring connections are secure.



Electronics Integration

Provide power to the Arduino and connect all modules properly. Verify the SDA and SCL connections for the I2C LCD and the data pin for the DHT11 sensor.



Program Upload

Write the Arduino program to read temperature and humidity data from the DHT11 sensor and upload the code using the Arduino IDE.



System Activation

The Arduino processes the sensor data and sends the temperature and humidity values to the LCD display for real-time monitoring.



Testing & Observation

Test the system under different environmental conditions and observe the displayed readings for accuracy and stability.



Final Operation

The weather station continuously monitors environmental temperature and humidity and displays the readings on the LCD screen.

Functioning Of The Device:

- The Arduino UNO will be programmed to read data from the DHT11 sensor and control the display system of the weather station.



- When the system powers on, the DHT11 sensor will detect the surrounding temperature and humidity of the environment.



- The sensor will send this environmental data to the Arduino UNO, where the microcontroller processes the readings.



- The processed temperature and humidity values will then be transmitted to the 16×2 LCD display through the I2C module.



- The LCD screen will continuously show real-time temperature and humidity readings, allowing users to monitor environmental conditions easily.

Thank You